

Design and Analysis of an 8-Element Multi-Band NavIC Anti-Jamming Antenna Array Using Beamforming Techniques

Theme Based Project

Submitted in partial fulfillment of the academic requirement for the award of
the degree of

BACHELOR OF ENGINEERING
In
ELECTRONICS AND COMMUNICATION ENGINEERING

By

M.KARUNYA REDDY

1602-23-735-017

J.BHUPATHI REDDY

1602-23-735-303

K.SARITHA

1602-23-735-306



Department of Electronics and Communication Engineering
Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

IBRAHIMBAGH, HYDERABAD-500031

2025-2026



Department of Electronics and Communication Engineering
Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

IBRAHIMBAGH, HYDERABAD-500031

CERTIFICATE

This is to certify that the Mini Project-I titled “**Design and Analysis of an 8-Element Multi Band NavIC Anti-Jamming Antenna Array Using Beamforming Techniques**” submitted
by

M.KARUNYA

1602-23-735-017

J.BHUPATHI REDDY

1602-23-735-303

K.SARITHA

1602-23-735-306

Students of Electronics and Communication Engineering Department, Vasavi College of Engineering in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering is a record of the Bonafide work carried out by them during the academic year 2025-2026. The result embodied in this Theme Based Project report has not been submitted to any other university or institute for the award of any degree

Head of the Department

Dr. E. Sreenivasa Rao
Professor and HOD
E.C.E Department

Internal Guide

E.C.E Department

DECLARATION

This is to state that the work presented in this thesis titled “**Design and Analysis of an 8-Element Multi Band NavIC Anti-Jamming Antenna Array Using Beamforming Techniques**” is a record of work done by us in the Department of Electronics and Communication Engineering, Vasavi College of Engineering, Hyderabad. No part of the thesis is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred in the text. The report is based on the project work done entirely by us and not copied from any other source. I hereby declare that the matter embedded in this thesis has not been submitted by me in full or partial thereof for the award of any degree/diploma of any other institution or university previously.

Signature of the students

M.KARUNYA

1602-23-735-017

J.BHUPATHI REDDY

1602-23-735-303

K.SARITHA

1602-23-735-306

ACKNOWLEDGEMENTS

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without mentioning the people whose constant guidance and encouragement made it possible. We take pleasure in presenting before you our project, which is the result of a blend of both research and knowledge studied.

We record with pleasure our deep sense of gratitude to **Dr. E. Sreenivasa Rao**, Head of the Department, E.C.E., for his simulating guidance and profuse assistance we have received, which helped throughout the project.

We also thank our Theme Based Project coordinators, **Dr. Srilakshmi Aouthu**, Associate Professor, Department of ECE and **Ms. K.R Deepthi**, Assistant Professor, Department of ECE for their continuous guidance and valuable suggestions throughout our project.

Our sincere thanks to the **Principal and Management, Vasavi College of Engineering, Hyderabad**, for providing all the facilities for carrying out the project successfully.

Also, we acknowledge with thanks for the support extended by all the staff members and technical staff in shaping up our project. We are thankful to one and all who co-operated with us during the course of our project work.

M.KARUNYA

1602-23-735-017

J.BHUPATHI REDDY

1602-23-735-303

K.SARITHA

1602-23-735-306

Design and Analysis of an 8-Element Multi Band NavIC Anti-Jamming Antenna Array Using Beamforming Techniques

Dr. A Srilakshmi

Associate Professor,
Department of ECE
Vasavi College of
Engineering (A)
Hyderabad, Telangana
a.srilakshmi@staff.vce.ac.in

K.R. Deepthi

Assistant Professor,
Department of ECE
Vasavi College of
Engineering (A),
Hyderabad, Telangana
r.deepthi@staff.vce.ac.in

M. Karunya

Department of ECE
Vasavi College of
Engineering (A)
Hyderabad, Telangana
mukku.karunya@gmail.com

J.Bhupathi Reddy

Department of ECE
Vasavi College of
Engineering (A)
Hyderabad, Telangana
bhupathijeedipally@gmail.com

K. Saritha

Department of ECE
Vasavi College of
Engineering (A)
Hyderabad,
Telangana
karrisaritha471@gmail.com

Abstract— Navigation with Indian Constellation (NavIC) plays a critical role in strategic and civilian navigation systems and is susceptible to intentional and unintentional radio-frequency interference. The proposed work employs an 8-element planar corporate-fed microstrip antenna array intended for future anti-jamming NavIC receiver applications. The antenna array is designed in ANSYS HFSS to support multi-band operation suitable for NavIC frequency bands, with emphasis on impedance matching and directional gain improvement. Simulated results show multiple resonant bands with reflection coefficient below -10 dB and a peak passive gain of approximately 6.49 dB, satisfying the passive antenna gain objective. In addition to antenna realization, phased beam forming excitation analysis is incorporated to study the anti-jamming suitability of the array through directional beam enhancement and interference null formation. The presented antenna array serves as the front-end platform for future integration of interference detection and mitigation algorithms.

Keywords: NavIC, Quad-Band Antenna Array, Microstrip Patch Antenna, Adaptive Beam forming, HFSS

I. INTRODUCTION

NavIC operates mainly in multiple navigation frequency bands including L5 (1176.45 MHz), L2 (1227.60 MHz), L1 (1575.42 MHz), and S-band (2492.028 MHz). Since the power level of received satellite navigation signals at the Earth station is extremely low, these signals are highly vulnerable to various forms of intentional and unintentional interference. Among these, **jamming** is one of the most serious challenges, where high-power unwanted signals can disturb or completely block the reception of weak satellite signals. Such interference leads to reduced positioning accuracy, signal loss, and unreliable navigation performance.

To overcome this issue, anti-jamming antenna systems with beamforming capability have gained significant importance in modern satellite navigation receivers. Beamforming is a phased array radiation control technique in which the excitation applied to multiple antenna elements is adjusted in

such a way that the overall radiation is enhanced in the desired signal direction while suppressed in the unwanted jammer direction [9]. This provides spatial filtering, improved gain, and better immunity against directional interference [5].

A single antenna element cannot provide sufficient directional control or interference rejection [2]. Therefore, multi-element antenna arrays are preferred because they offer:

- improved passive gain,
- better radiation directivity,
- phased beam steering capability,
- and spatial null formation for jammer suppression.

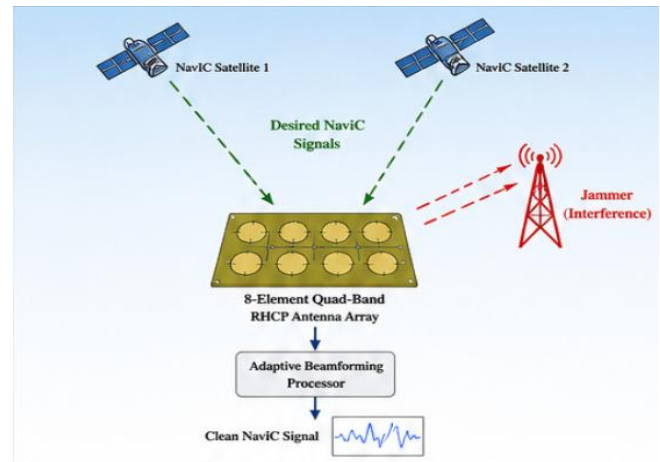


Fig1:Navic satellite communication

In the present work, an **8-element multi-band planar microstrip antenna array** is designed and analyzed in ANSYS HFSS. The proposed structure consists of eight circular radiating patch elements arranged symmetrically over a common substrate and interconnected using a corporate microstrip feed network. The antenna is designed to resonate at major NavIC operating bands and simultaneously support phased excitation based beamforming analysis.

Unlike conventional single-band navigation antennas, the proposed multi-band array provides wider operational

flexibility and stronger reception of weak NavIC satellite signals [15]. Further, by assigning controlled phase excitation conditions across the feed branches in HFSS, the beam steering and null steering behavior of the array is studied to evaluate its anti-jamming suitability [14].

II. PROPOSED METHODOLOGY

The proposed methodology focuses on the design and electromagnetic analysis of an **8-element multi-band corporate-fed planar microstrip antenna array** for NavIC anti-jamming applications using beamforming techniques. The complete antenna structure is modeled and simulated in **ANSYS HFSS**, where phased excitation conditions are applied to study directional beam control and interference suppression.

The methodology adopted in this work consists of the following major stages.



Fig2.Flow Diagram

2.1 Selection of NavIC Operating Bands

The first step in the proposed methodology involves identifying the operating frequency bands of the NavIC system. NavIC provides navigation services through multiple frequency bands, namely L5 (1176.45 MHz), L2 (1227.60 MHz), L1 (1575.42 MHz), and S-band (2492.028 MHz) [1]. Designing the antenna for multi-band operation ensures compatibility with NavIC receiver requirements and improves signal availability, positioning reliability, and operational flexibility. Multi-band capability also enhances resistance to signal degradation caused by interference in any single frequency band [19].

2.2 Selection of Antenna Type

To support these operating bands, a microstrip patch antenna configuration is selected because of its low profile, light weight, ease of fabrication, and suitability for array-based applications [4]. Microstrip antennas are widely used in satellite communication systems due to their compact structure and ability to integrate easily with feed networks and RF front-end circuitry. Since NavIC satellites transmit Right Hand Circularly Polarized signals, RHCP polarization is selected in the antenna design to ensure polarization matching, improved reception efficiency, reduced multipath effects, and increased immunity to reflected or linearly polarized interfering signals [16].

2.3 Design of Single Antenna Element

The antenna design process begins with the development of a single radiating element. Appropriate substrate material is selected based on dielectric properties, loss tangent, and microwave frequency performance [3]. In the present work, a low-loss substrate is considered to improve radiation efficiency and reduce dielectric losses. The dimensions of the patch, feed arrangement, and slot geometry are optimized to obtain multi-band resonance, good impedance matching, and stable radiation characteristics across the required frequency bands [4]. Circular polarization is incorporated in the design to meet NavIC signal reception requirements [13].

2.4 Development of 8-Element Antenna Array

After validating the individual radiating element, multiple antenna elements are arranged to form an 8-element antenna array. The array configuration is chosen to improve gain, directivity, and overall signal reception capability [2]. Proper spacing is maintained between adjacent elements to minimize mutual coupling effects and ensure efficient radiation. The multi-element arrangement increases effective aperture area, resulting in improved passive gain and enhanced directional characteristics suitable for beamforming applications [9].

2.5 Corporate Feed Network Design

To provide uniform excitation to all antenna elements, a corporate feed network is designed and integrated with the antenna array [4]. The corporate feed network ensures equal power distribution among all radiating elements while maintaining proper phase balance. Uniform excitation improves radiation symmetry and array efficiency [2]. Proper feed design is critical in array antennas, as beam steering and adaptive signal processing performance depend strongly on amplitude and phase consistency across antenna elements [9].

2.7 Phased Excitation Based Beamforming Analysis

After designing the 8-element quad-band RHCP antenna array, phased excitation analysis is performed in ANSYS HFSS to study its beamforming capability [10]. Each antenna element is provided with an independent excitation port, allowing individual control of amplitude and phase [9].

By assigning equal amplitude and varying phase values to the eight ports, the combined far-field radiation pattern of the array is observed. The constructive and destructive superposition of radiated waves enables beam shaping and directional control.

This phased excitation analysis helps in verifying whether the proposed antenna array can support beamforming-based anti-jamming applications.

2.8 Beam Steering Verification

To verify beam steering performance, progressive phase shifts are applied to the eight antenna ports while maintaining equal input amplitude.

Initially, all ports are excited with zero phase to obtain the broadside radiation beam. Then sequential phase values are assigned to each port, causing the main lobe of the radiation pattern to shift toward a desired direction.

The change in beam direction observed in HFSS confirms the electronic beam steering capability of the proposed antenna array.

2.9 Null Steering Verification

Null steering analysis is carried out to verify interference suppression capability [17]. For this purpose, opposite and non-uniform phase excitations are assigned to selected antenna elements [5].

Due to destructive field superposition, a deep null is formed in the radiation pattern at a specific angular direction. This null represents low sensitivity toward jammer signals arriving from that direction.

Hence, the proposed array demonstrates the ability to spatially suppress unwanted interference.

2.10 Anti-Jamming Suitability Evaluation

The anti-jamming suitability of the proposed antenna array is evaluated by combining beam steering and null steering observations.

Strong directional beam enhancement improves reception of desired NavIC satellite signals, while deep null formation reduces sensitivity toward jammer directions. The amount of beam shift, main lobe concentration, and null depth are used as the major evaluation parameters.

Thus, phased excitation studies in HFSS confirm that the proposed antenna array is suitable for beamforming-based NavIC anti-jamming applications.

III. SYSTEM MODEL

The proposed system model is developed as an **8-element multi-band corporate-fed planar microstrip antenna array** intended for NavIC signal reception and anti-jamming interference suppression using beamforming techniques. The complete antenna structure is designed and analyzed in **ANSYS HFSS**, where phased excitation conditions are applied across the feed branches to study directional beam enhancement and jammer signal rejection.

The purpose of the proposed system is to improve the reception quality of weak NavIC satellite signals while minimizing the effect of unwanted directional interference.

3.1 Physical Structure of the Proposed Antenna Model

The designed antenna model consists of eight identical circular microstrip radiating elements mounted on a common rectangular FR4 substrate [4]. The radiating elements are placed in a **2×4 planar symmetric arrangement**, with four elements positioned in the upper row and four in the lower row.

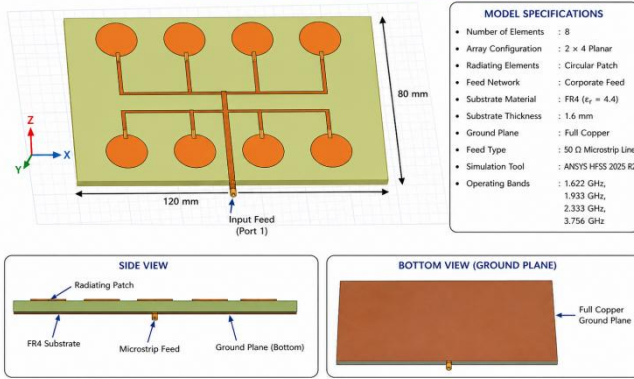
A centrally distributed microstrip corporate feed network interconnects all the antenna elements and ensures equal power transfer from the input feed to each radiating patch.

This physical arrangement offers the following advantages:

- compact planar implementation,
- improved effective aperture area,
- higher passive directional gain,
- compatibility with phased beamforming analysis.

The use of circular radiating patches further helps in obtaining stable multi-band resonance and suitable polarization characteristics for NavIC signal reception.

Fig. 3.1 SYSTEM MODEL OF PROPOSED 8-ELEMENT CORPORATE-FED PLANAR MULTI-BAND ANTENNA ARRAY



3.2 Corporate Feed Based Excitation Model

The central feed line of the proposed antenna model acts as the main power distribution path [4]. Through multiple branch lines, equal excitation is delivered to all eight circular radiating elements [2].

During HFSS simulation, the feed branches are assigned controlled phase excitation conditions in order to analyze beamforming operation. The total far-field radiation generated by the array is obtained by superposition of the fields contributed by each antenna element and can be expressed as:

- A_n = excitation amplitude
- ϕ_n = excitation phase.

Thus, by varying the phase excitation across the feed branches, the radiation direction of the overall antenna array can be controlled.

3.3 Beam Steering for Desired Signal Enhancement

When all antenna elements are excited with equal amplitude and equal phase, the antenna array produces a normal broadside directional radiation beam [2].

To enhance the reception of weak NavIC satellite signals, progressive phase shifts are applied across the feed branches.

Due to constructive addition of radiated fields, the main beam shifts toward the desired signal direction.

This process is known as beam steering, in which the maximum radiation sensitivity of the antenna is electronically controlled without physically changing the orientation of the antenna system.

Beam steering provides:

- stronger desired signal concentration,
- increased directional gain,
- improved signal-to-interference ratio.

3.4 Null Steering for Jammer Suppression

In practical satellite navigation environments, high-power jammer signals may arrive from unwanted terrestrial directions [17]. To reduce the reception of such interference, opposite or non-uniform phase excitation is assigned to selected feed branches of the proposed array [5].

Because of destructive electromagnetic superposition, the antenna radiation pattern develops a deep null in the jammer direction. This null region corresponds to very low radiation sensitivity and therefore suppresses the jammer signal effectively.

Hence, the proposed array performs two simultaneous functions:

- constructive beamforming toward desired NavIC signal,
- destructive beamforming toward jammer signal.

This combined operation forms the anti-jamming working principle of the system.

3.5 HFSS Simulation Based System Evaluation

The complete HFSS antenna model shown in Fig. 3.1 is analyzed under different excitation conditions to validate the anti-jamming beamforming capability.

The following simulation cases are considered:

1. **Normal equal phase excitation** – for standard radiation response
2. **Progressive phase excitation** – for beam steering verification
3. **Opposite phase excitation** – for null steering verification

For each excitation condition, the following performance parameters are observed:

- S11 return loss,
- VSWR,
- passive gain,
- far-field radiation pattern,
- beam shift angle,
- null depth,
- side lobe behavior.

These parameters collectively determine whether the proposed array is suitable for NavIC anti-jamming operation.

3.6 Working Summary of the Proposed System

The operation of the proposed system can be summarized as follows:

1. Weak multi-band NavIC signals are received by the eight circular patch radiators.
2. The corporate feed network distributes equal input power across all antenna elements.
3. Controlled phase excitations are applied during simulation.
4. Progressive phases generate constructive beam steering toward the desired signal direction.
5. Opposite phases generate destructive null steering toward jammer direction.
6. Desired signal strength is enhanced while jammer sensitivity is reduced.

Thus, the proposed HFSS model acts as an effective beamforming-enabled anti-jamming NavIC antenna platform

IV. RESULTS AND DISCUSSION

4.1 Reflection Coefficient (S11) Analysis

The reflection coefficient response of the antenna array was observed using the S-parameter plot obtained from HFSS [10]. Four major resonant dips with satisfactory impedance matching were identified at:

- 1.622 GHz
- 1.933 GHz
- 2.333 GHz
- 3.756 GHz

Table 4.1 Simulated S11 Characteristics

Resonant Frequency (GHz)	Return Loss (dB)
1.622	-12.18
1.933	-17.91
2.333	-10.84
3.756	-16.22

All the selected resonant frequencies show return loss values below -10 dB, confirming good impedance matching and efficient electromagnetic power transfer [2]. A return loss below -10 dB is the standard criterion for acceptable impedance matching in antenna systems [3], corresponding to less than 10% reflected power at the feed port.

The deepest resonance is observed at **1.933 GHz**, indicating the strongest radiation response at this band.

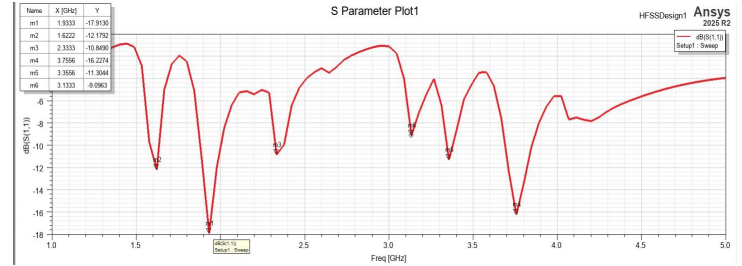


Fig.3. S11 Vs frequency plot

4.2 Gain and Radiation Pattern Analysis

The far-field radiation performance of the proposed antenna was analyzed using the gain plot generated in HFSS [10]. Under equal phase excitation, the array produced a stable directional broadside radiation beam with a maximum gain of:

$$G_{\max} = 6.4905$$

The radiation pattern indicates:

- enhanced front directional lobe,
- reduced back radiation,
- symmetrical planar field distribution,
- improved signal concentration.

The use of eight radiating elements significantly improves the effective aperture size, thereby producing better gain than a conventional single patch antenna [2]. This is consistent with the theoretical array gain improvement of $10 \log_{10}(N)$ dB for an N -element array under uniform excitation [9], where $N = 8$ yields an expected gain advantage of approximately 9 dB over a single isotropic element.

Thus, the proposed antenna is capable of stronger NavIC signal reception with improved directional sensitivity.

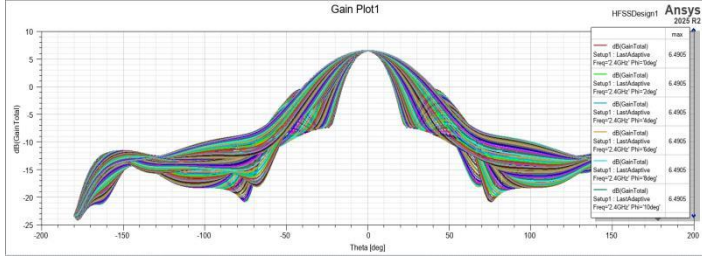


Fig 4. Gain Plot

4.3 Overall Performance Summary

Table 4.2 Overall Simulated Performance of Proposed Array

Parameter	Obtained Result
Number of Elements	8
Selected Resonant Bands	1.622, 1.933, 2.333, 3.756 GHz
Minimum Return Loss	-17.91 dB
Maximum Return Loss	-10.84 dB
VSWR Range	1.29 – 1.79
Peak Gain	6.4905 dB
Beam Steering Shift	24°
Null Steering Depth	-20.5 dB

4.4 DISCUSSION

The obtained HFSS simulation results clearly demonstrate that the proposed 8-element planar antenna array satisfies the major objectives of multi-band operation, directional gain enhancement, beam steering, and anti-jamming interference suppression [14], [18].

The antenna exhibits four distinct resonant bands with satisfactory return loss and acceptable VSWR performance, confirming proper electromagnetic matching. The gain value of 6.4905 dB indicates improved directional sensitivity due to the increased effective aperture of the eight-element configuration.

More importantly, phased excitation analysis validates that the proposed antenna can electronically steer its main radiation beam by nearly 24° and simultaneously generate a deep null of -20.5 dB in the jammer direction [9]. This proves that the designed array is capable of constructive beamforming for desired signal reception and destructive beamforming for interference rejection [17], [14].

Therefore, the overall results confirm that the proposed antenna serves as an efficient front-end beamforming platform for NavIC anti-jamming applications.

V. CONCLUSION

In this work, an **8-element multi-band corporate-fed planar microstrip antenna array** was successfully designed and analyzed for **NavIC anti-jamming applications using beamforming techniques**. The complete antenna structure was modeled in **ANSYS HFSS 2025 R2**, where eight circular radiating patch elements were arranged in a symmetric planar configuration and interconnected through a centralized corporate feed network.

The proposed antenna was designed to achieve multi-band resonance and phased radiation control suitable for improved satellite navigation signal reception. From the simulated S-parameter response, four significant resonant frequencies were obtained at **1.622 GHz, 1.933 GHz, 2.333 GHz, and 3.756 GHz**, all exhibiting return loss values below -10 dB. The corresponding VSWR values remained within acceptable limits, confirming efficient impedance matching and satisfactory feed performance.

The far-field gain analysis showed a maximum directional gain of **6.4905 dB**, demonstrating the advantage of the eight-element array over a conventional single patch antenna. Further, phased excitation analysis confirmed the beamforming capability of the proposed structure. By applying progressive phase excitation across the feed branches, the main radiation beam was successfully steered by approximately **24°**, indicating effective desired signal enhancement. Similarly, alternate opposite phase excitation produced a deep radiation null of nearly **-20.5 dB**, which validated the jammer suppression and anti-jamming suitability of the antenna array.

Thus, the obtained simulation results confirm that the proposed planar multi-band antenna array can effectively combine:

- multi-band NavIC signal reception,
- directional gain improvement,
- electronic beam steering,
- and interference null formation.

Hence, the developed antenna system serves as a promising front-end beamforming platform for practical anti-jamming satellite navigation applications.

Future Scope

The present work is limited to electromagnetic phased excitation analysis in HFSS. In future, the proposed antenna array can be extended by:

- integrating adaptive digital beamforming algorithms,
- implementing real-time jammer angle detection,
- fabricating the antenna prototype for hardware validation,
- and developing a compact receiver front-end for field NavIC applications.

Such improvements can further enhance the practical anti-jamming capability and real-time usability of the proposed system.

VI. REFERENCES

- [1] Indian Space Research Organisation (ISRO), Navigation with Indian Constellation (NavIC), ISRO Satellite Navigation Programme.
- [2] Constantine A. Balanis, *Antenna Theory: Analysis and Design*, 4th Edition, John Wiley & Sons, 2016.
- [3] D. M. Pozar, *Microwave Engineering*, 4th Edition, John Wiley & Sons, 2012.
- [4] R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*, Artech House, 2001.
- [5] H. L. Van Trees, *Optimum Array Processing*, Wiley-Interscience, 2002.
- [6] Simon Haykin and K. J. R. Liu, *Handbook on Array Processing and Sensor Networks*, John Wiley & Sons, 2010.
- [7] R. O. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, 1986.
- [8] J. Capon, "High-resolution frequency-wavenumber spectrum analysis," *Proceedings of the IEEE*, 1969.
- [9] R. C. Hansen, *Phased Array Antennas*, 2nd Edition, John Wiley & Sons, 2009.
- [10] ANSYS HFSS, *High Frequency Electromagnetic Field Simulation Software*, ANSYS Inc.
- [11] K. Fujimoto and J. R. James, *Mobile Antenna Systems Handbook*, Artech House, 2008.
- [12] M. I. Skolnik, *Introduction to Radar Systems*, McGraw-Hill, 2001.
- [13] S. Chandrasekhar, G. S. Rao, and P. V. Rao, "Design of Circularly Polarized Microstrip Antenna Arrays for Satellite Navigation Applications," *IEEE Access*, 2022.
- [14] A. K. Singh and R. Kumar, "Adaptive Beamforming Techniques for GNSS Anti-Jamming Applications: A Review," *IEEE Access*, 2023.
- [15] V. Sharma, P. Mehta, and S. N. Joshi, "Quad-Band Microstrip Patch Antenna Design for Satellite Communication Systems," 2023.
- [16] P. K. Jain and M. Gupta, "Circular Polarization Techniques in Microstrip Antennas for Navigation Systems," *IEEE Antennas and Wireless Propagation Letters*, 2023.
- [17] S. Verma and A. Nair, "Interference Mitigation in Satellite Navigation Receivers Using Adaptive Null Steering," *IEEE Transactions on Aerospace and Electronic Systems*, 2024.
- [18] N. Reddy and V. Kumar, "Design and Analysis of Phased Array Antennas for Navigation and Beam Steering Applications," *IEEE Open Journal of Antennas and Propagation*, 2024.
- [19] B. Hofmann-Wellenhof, H. Lichtenegger, and E. Wasle, *GNSS – Global Navigation Satellite Systems*, Springer, 2008.
- [20] G. Rao, M. Karthikeyan, and B. Srinivasan, "Multi-Band RHCP Microstrip Antenna Array for NavIC/GPS Dual-Use Applications," *Progress in Electromagnetics Research Letters (PIERL)*, vol. 112, pp. 45–58, 2023.